



B(2nd Sm.)-Computer Sc.-MDC/CC-2/CCP

2024

COMPUTER SCIENCE — MDC

Paper : CC-2

(Problem Solving Using C)

Full Marks : 75

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer *question no. 1* and
any three from *Section-A* and *any five* from *Section-B*.

1. Answer *any five* questions :

2/5

- (a) Name the four basic data types in C. Also name the four data-type qualifiers.
- (b) Define relational and logical operators with suitable example.
- (c) Explain entry control and exit control loop with suitable example.
- (d) Can we use string variable in switch test condition? Explain with example.
- (e) What is the use of Null pointer?
- (f) Define an array. Write the syntax for multidimensional array in C.
- (g) What is the use of fclose() function in C? Explain with example.
- (h) What is the purpose and prototype of the function 'strcpy'?

Section - A

2. Enumerate the difference between 'if-else' and 'switch-case' statements with suitable example. 5
3. Write a C program to compare two strings without using strcmp () function. 5
4. Write a C program to display all odd numbers between 51 and 90. Also find the sum of numbers. 5
5. What are the various storage classes in C? What are their uses? Explain with example. 5
6. What are file accessing modes in C? Explain any two modes with suitable example. 5

Please Turn Over

Section - B

7. (a) How can initial value be assigned to variable during type declaration?
(b) How is a string assigned to a one dimensional array in C?
(c) What is global variable in C and what are its uses? Explain with suitable example. 3+3+4
8. (a) What are increment and decrement operators? Explain their uses with suitable example.
(b) How does "sizeof" operator work? Explain.
(c) Write a C program to print the first 'n' Fibonacci numbers. 'n' will be taken as input from the user. 2+3+5
9. (a) State the difference between call-by-value and call-by-reference. Illustrate with suitable example.
(b) Write a C program to check whether a year is leap year or not. Print 1 for leap year otherwise print 0. 5+5
10. (a) What is the function of '# define' statement in C?
(b) Differentiate between structure and union.
(c) Write a C program to generate the following pattern and take number of lines as input. (In this example n = 3)
- ★
★ ★ ★
★ ★ ★ ★ ★
- 2+3+5
11. (a) What is the difference between calloc() and malloc() function? Explain with example.
(b) Describe two bit-wise operator in brief.
(c) Write a program in C to check whether a number is Armstrong or not. 3+2+5
12. (a) What is recursion? Give an example to explain it.
(b) What is the use of '\t'?
(c) What is nested structure? Explain with suitable example. 3+2+5
13. (a) Can you recursively call main() from main()? Justify.
(b) Write a program in C that calculates 'a raised to the power b' without using pow() function.
(c) Write a program in C to print the content of the file 'file1.txt'. 2+3+5
14. (a) Write your own function to simulate the working of the in-built strlen() function.
(b) How can you pass 2-D arrays to functions? Discuss the rules. 5+(3+2)